Computational Linear Algebra

Project Report

on

Arnoldi Iteration: Krylov Subspaces, GMRES and Ritz Value Approximation

by

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Abstract

This report studies the Arnoldi iteration and its connection with Krylov subspaces, eigenvalue approximation and the Generalized Minimal Residual (GMRES) method. We first develop the Arnoldi process and interpret it as an orthogonal projection of a linear operator onto a Krylov subspace. We then examine how the eigenvalues of the resulting Hessenberg matrix, known as Ritz values, approximate the eigenvalues of the original matrix. The Ritz residual and the Galerkin condition are used to understand the quality of these approximations, with particular attention to the difference between normal and nonnormal matrices.

The report also develops GMRES from its least-squares formulation and studies its interpretation as a polynomial minimisation problem. This viewpoint helps explain how the location of eigenvalues can influence convergence, and why eigenvalues alone may not fully describe the behaviour of GMRES for nonnormal matrices. Numerical experiments illustrate these ideas through comparisons of normal and nonnormal matrices, as well as matrices with disk-shaped and rose-shaped eigenvalue distributions. These examples provide a computational perspective on the theoretical properties of Arnoldi iteration and GMRES.

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1 Arnoldi Iteration

Arnoldi iteration orthogonalises the vectors $\{q_1, Aq_1, Aq_2, \dots\}$. It uses the same process of orthogonalization as that of Gram–Schmidt. (Here A is an $m \times m$ matrix.) We start with a normalized vector q_1, $\|q_1\| = 1$.

Step 1

We calculate the first inner product

$$h_{11} = \langle q_1, Aq_1 \rangle \qquad \text{(component of } Aq_1 \text{ in the direction of } q_1\text{)}.$$

Now we subtract the projection:

$$r_1 = Aq_1 - h_{11}q_1.$$

We calculate the norm of the residual

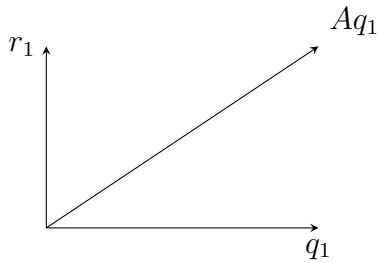
$$h_{21} = \|r_1\| = \|Aq_1 - h_{11}q_1\|.$$

If $h_{21} \neq 0$, we normalize

$$q_2 = \frac{r_1}{h_{21}}.$$

Hence

$$Aq_1 = \underbrace{h_{11}q_1}_{\text{component in direction of } q_1} + \underbrace{h_{21}q_2}_{\text{component in direction of } q_2}.$$



Let $Q_2 = \begin{bmatrix} q_1 & q_2 \end{bmatrix}$ and $\tilde{H}_1 = \begin{bmatrix} h_{11} \\ h_{21} \end{bmatrix}$. Then

$$\boxed{Aq_1 = Q_2\tilde{H}_1}$$

Step 2

We process Aq_2. Similarly we have

$$Aq_2 = h_{12}q_1 + h_{22}q_2 + h_{32}q_3,$$

where

$$h_{12} = \underbrace{\langle q_1, Aq_2 \rangle}_{\text{component in the direction of } q_1}, \qquad h_{22} = \underbrace{\langle q_2, Aq_2 \rangle}_{\text{component in the direction of } q_2},$$

and h_{32} is the norm of the residual

$$r_2 = Aq_2 - h_{12}q_1 - h_{22}q_2.$$

Thus $Aq_2 = h_{12}q_1 + h_{22}q_2 + h_{32}q_3$

$$\implies \quad AQ_2 = Q_3\tilde{H}_2, \qquad \text{where} \quad \tilde{H}_2 = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \\ 0 & h_{32} \end{bmatrix}_{3\times 2}.$$

Thus after n iterations we have

$$\boxed{AQ_n = Q_{n+1}\tilde{H}_n} \tag{1}$$

$$\tilde{H}_n = \begin{bmatrix} h_{11} & h_{12} & \cdots & h_{1n} \\ h_{21} & h_{22} & \cdots & h_{2n} \\ & \ddots & \ddots & \vdots \\ & & h_{n,n-1} & h_{nn} \\ & & & h_{n+1,n} \end{bmatrix}_{(n+1)\times n}$$

Thus,

$$\begin{bmatrix} \\ A \\ \\ \end{bmatrix} \begin{bmatrix} | & & | \\ q_1 & \cdots & q_n \\ | & & | \end{bmatrix} = \begin{bmatrix} | & & | & | \\ q_1 & \cdots & q_n & q_{n+1} \\ | & & | & | \end{bmatrix} \begin{bmatrix} h_{11} & \cdots & h_{1n} \\ h_{21} & \ddots & \vdots \\ & \ddots & h_{nn} \\ & & h_{n+1,n} \end{bmatrix}$$

$$\boxed{Aq_n = h_{1n}q_1 + \cdots + h_{nn}q_n + h_{n+1,n}q_{n+1}}$$

Why is H upper Hessenberg? In Arnoldi, Aq_j is represented using $q_1, \ldots, q_{j+1}$. The extra basis vector q_{j+1} introduces an extra nonzero entry immediately below the diagonal.

Algorithm 1 Arnoldi Iteration

Require: $A \in \mathbb{C}^{m\times m}$, $q_1 \in \mathbb{C}^m$ with $\|q_1\|_2 = 1$, number of steps n
Ensure: $Q_{n+1} = [q_1, \ldots, q_{n+1}]$, $\tilde{H}_n \in \mathbb{C}^{(n+1)\times n}$

```
for j = 1, 2, ..., n do
    v ← A q_j
    for i = 1, 2, ..., j do
        h_ij ← q_i^* v
        v ← v − h_ij q_i
    end for
    h_{j+1,j} ← ||v||_2
    if h_{j+1,j} = 0 then
        stop
    end if
    q_{j+1} ← v / h_{j+1,j}
end for
```

Remark 1.1. Algorithm 1 is the modified Gram–Schmidt form of Arnoldi: each h_{ij} is computed against the already-updated v, which is numerically more stable. The notes above use the classical form, where all $h_{ij} = q_i^* A q_j$ are computed against Aq_j directly. In exact arithmetic the two produce identical Q_{n+1} and $\tilde{H}_n$.

2 Another Interpretation of Arnoldi Process: Projection onto Krylov Spaces

$$K_n(A, q_1) = \langle q_1, Aq_1, A^2q_1, \ldots, A^{n-1}q_1 \rangle.$$

Instead of working with potentially huge matrix A directly, we represent its action on this subspace using a much smaller matrix H_n.

$$\boxed{AQ_n = Q_{n+1}\tilde{H}_n}$$

Let $A \in \mathbb{C}^{m\times m}$, and let $Q_n = \begin{bmatrix} q_1 & \cdots & q_n \end{bmatrix} \in \mathbb{C}^{m\times n}$ have orthogonal columns. So

$$\boxed{Q_n^* Q_n = I_n}$$

We define $K_n = \text{Range}(Q_n) = \langle q_1, q_2, \ldots, q_n \rangle$. We define $P_n : \mathbb{C}^m \to \mathbb{C}^m$ by

$$\boxed{P_n = Q_n Q_n^*}$$

Remark 2.1. If $h_{j+1,j} \neq 0$ for $j = 1, \ldots, n$ (no breakdown), then

$$K_n = \text{Range}(Q_n) = \langle q_1, \ldots, q_n \rangle = K_n(A, q_1).$$

Indeed, by construction $q_{j+1} \in \langle q_1, \ldots, q_j, Aq_j \rangle$, so by induction $q_j \in K_j(A, q_1)$ for all j. Since $q_1, \ldots, q_n$ are orthonormal, hence linearly independent, they span the n-dimensional space $K_n(A, q_1)$.

Proposition 2.1. *$P_n = Q_n Q_n^*$ is an orthogonal projector onto the subspace K_n.*

Proof.

$$P_n^2 = (Q_n Q_n^*)(Q_n Q_n^*) = Q_n (Q_n^* Q_n) Q_n^* = Q_n Q_n^* = P_n, \qquad P_n^* = P_n.$$

Also

$$P_n x = Q_n Q_n^* x = \sum_{j=1}^{n} q_j \, (q_j^* x) \in K_n,$$

i.e.

$$\underbrace{\begin{bmatrix} q_1 & q_2 & \cdots & q_n \end{bmatrix}}_{m\times n} \underbrace{\begin{bmatrix} q_1^* \\ q_2^* \\ \vdots \\ q_n^* \end{bmatrix}}_{n\times m} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix}}_{m\times 1}.$$

Also

$$Q_n Q_n^*(x - P_n x) = Q_n Q_n^* x - Q_n (Q_n^* Q_n) Q_n^* x = Q_n Q_n^* x - Q_n Q_n^* x = 0.$$

Multiplying on the left by Q_n^* gives $Q_n^*(x - P_n x) = 0$. So,

$$\boxed{x - P_n x \perp K_n}$$

Thus we have proved that $P_n = Q_n Q_n^*$ is an orthogonal projector onto the subspace K_n. □

In general $A(K_n) \not\subseteq K_n$. So we need an operator such that post-composing A restricted to K_n lies inside the subspace K_n.

$$A : \mathbb{C}^m \to \mathbb{C}^m, \qquad A|_{K_n} : K_n \to \mathbb{C}^m$$

$$\begin{array}{ccc} K_n & \xrightarrow{A|_{K_n}} & \mathbb{C}^m \\ & \searrow^{T_n} & \downarrow{\scriptstyle P_n} \\ & & K_n \end{array}$$

$$\boxed{T_n := P_n \circ A|_{K_n}}$$

Thus we have $T_n = Q_n Q_n^* A|_{K_n} : K_n \to K_n$.

Now we will find the matrix of T_n in the basis of Q_n. We will see that this matrix is nothing but the Hessenberg matrix H_n. We define

$$\varphi_n : \mathbb{C}^n \to K_n, \quad \varphi_n(y) = Q_n y, \qquad \varphi_n^{-1}(x) = Q_n^* x, \ x \in K_n.$$

$$\begin{aligned} [T_n]_{Q_n} &= \varphi_n^{-1}\, T_n\, \varphi_n \\ &= Q_n^* \left(P_n A|_{K_n}\right) Q_n \\ &= Q_n^* Q_n Q_n^* A|_{K_n} Q_n \\ &= I_n\, Q_n^* A|_{K_n} Q_n \\ &= Q_n^* A Q_n \end{aligned}$$

$$\boxed{H_n = Q_n^* A Q_n}$$

Thus H_n is the matrix of $T_n = P_n A|_{K_n}$ in the orthonormal basis Q_n.

Now we know the Arnoldi relation

$$\boxed{AQ_n = Q_{n+1}\tilde{H}_n} \tag{2}$$

where $Q_{n+1} = [q_1, \ldots, q_{n+1}]$, $\tilde{H}_n \in \mathbb{C}^{(n+1)\times n}$. Multiplying on the left side of (2) by Q_n^* we get

$$Q_n^* A Q_n = Q_n^* Q_{n+1} \tilde{H}_n.$$

Now

$$Q_n^* \begin{bmatrix} Q_n & q_{n+1} \end{bmatrix} = \begin{bmatrix} I_n & 0 \end{bmatrix} \qquad [\because\ Q_n^* Q_n = I_n,\ \langle q_{n+1}, q_j \rangle = 0 \ \ \forall j \in [1, n]]$$

$$\implies \quad H_n = \begin{bmatrix} I_n & 0 \end{bmatrix} \tilde{H}_n.$$

H_n is $\tilde{H}_n$ with the last row deleted. So the Hessenberg matrix from Arnoldi is exactly the same matrix of this operator.

3 Computing Eigenvalues by the Arnoldi Iteration

The eigenvalues of H_n are called *Ritz values*.

$$\boxed{H_n y = \theta y}$$

We know already that

$$\boxed{AQ_n = Q_n H_n + h_{n+1,n}\, q_{n+1} e_n^*} \tag{3}$$

Now if $H_n y = \theta y$ with $\|y\|_2 = 1$, define the *Ritz vector*

$$\boxed{u = Q_n y} \qquad \text{so } \|u\|_2 = 1.$$

From equation (3) we get (multiplying by y on both sides)

$$\boxed{Au - \theta u = h_{n+1,n}\, q_{n+1}\, e_n^* y} \tag{4}$$

Here the Ritz residual satisfies

$$\|Au - \theta u\|_2 = |h_{n+1,n}|\, |e_n^* y|.$$

A small residual means (θ, u) is an approximate eigenpair.

Remark 3.1 (Breakdown). If $h_{n+1,n} = 0$, then (3) becomes $AQ_n = Q_n H_n$, so K_n is A-invariant. By (4) the residual is then zero for every Ritz pair, i.e. every Ritz pair (θ, u) is an exact eigenpair of A.

Proposition 3.1 (Galerkin Condition)**.** *$Q_n^*(Au - \theta u) = 0$, i.e. the residual is orthogonal to $K_n(A, b)$, where*

$$K_n(A, b) = \langle b, Ab, A^2 b, \ldots, A^{n-1} b \rangle,$$

with $q_1 = b/\|b\|_2$ (so $K_n(A, b) = K_n(A, q_1) = \text{Range}(Q_n)$ by Remark 2.1, assuming no breakdown).

Proof. From equation (4) we get that $Au - \theta u$ is a multiple of q_{n+1}. We know that $q_{n+1} \perp q_1, q_2, \ldots, q_n$. Thus

$$\boxed{Au - \theta u \perp K_n(A, b)}$$

□

This Galerkin condition tells us that the error $r = Au - \theta u$ in the eigenvalue equation is orthogonal to the entire search space $\langle q_1, q_2, \ldots, q_n \rangle$.

Proposition 3.2. *Ritz values lie in the numerical range $W(A) = \{x^* A x : \ \|x\|_2 = 1\}$.*

Proof. The proposition claims that $\theta = u^* A u \in W(A)$. We already know that

$$Au - \theta u = r.$$

Multiply by u^*:

$$u^* Au - \theta u^* u = u^* r.$$

Also $u^* r = 0$ by the previous proposition. So

$$u^* Au - \theta u^* u = 0 \implies u^* Au = \theta u^* u \implies \theta = u^* Au$$

[if we assume $\|u\|_2 = 1$]. □

Proposition 3.3. *If A is normal and $\|Au - \theta u\|_2 \le \varepsilon$, $\|u\|_2 = 1$, then*

$$\boxed{d(\theta, \Lambda(A)) \le \varepsilon}$$

where $\Lambda(A)$ is the spectrum of A.

Proof. Since A is normal, using the spectral decomposition we get

$$A = V\Lambda V^*,$$

where V is unitary and $\Lambda = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_m)$. Define $z = V^*u$. Then $\|z\|_2 = \|V^*u\|_2 = 1$ [$\because V^*$ is unitary]. We have

$$Au - \theta u = V\Lambda V^*u - \theta u.$$

Since $u = Vz$,

$$Au - \theta u = V(\Lambda - \theta I)z, \qquad \Lambda - \theta I = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_m \end{bmatrix} - \begin{bmatrix} \theta & & \\ & \ddots & \\ & & \theta \end{bmatrix}.$$

Hence

$$\|Au - \theta u\|_2 = \|(\Lambda - \theta I)z\|_2 = \Big(\sum_{j=1}^{m} |\lambda_j - \theta|^2 |z_j|^2\Big)^{1/2}. \tag{5}$$

Let

$$d = d(\theta, \Lambda(A)) = \inf_{\lambda \in \Lambda(A)} |\lambda - \theta| = \min_{j \in [1,m]} |\lambda_j - \theta|.$$

Thus from (5) we get

$$\|Au - \theta u\|_2^2 = \sum_{j=1}^{m} |\lambda_j - \theta|^2 |z_j|^2 \;\geq\; d^2 \sum_{j=1}^{m} |z_j|^2 = d^2$$

$$\implies \quad d \leq \|Au - \theta u\|_2 \leq \varepsilon.$$

□

Remark 3.2. For an Arnoldi Ritz pair, we have

$$\|Au - \theta u\|_2 = |h_{n+1,n}|\,|e_n^* y|.$$

Therefore if A is normal

$$\boxed{d(\theta, \Lambda(A)) \leq |h_{n+1,n}|\,|e_n^* y|}$$

It tells that if the Ritz residual is small, then the Ritz value is guaranteed to be close to an actual eigenvalue. But for nonnormal cases, Ritz values can have small residuals while still being far from the spectrum.

4 Numerical Experiment: Normal versus Nonnormal Matrices

We compare a normal and a nonnormal matrix of size $m \times m$.

$$A_{\text{normal}} == \begin{bmatrix} 1 & & & \\ & 2 & & \\ & & \ddots & \\ & & & 30 \end{bmatrix}_{30\times 30}$$

$$A_{\text{nonnormal}} = \begin{bmatrix} 0 & 100 & 0 & \cdots & 0 \\ 0 & 0 & 100 & \ddots & \vdots \\ \vdots & & \ddots & \ddots & 0 \\ 0 & \cdots & & 0 & 100 \\ 0 & \cdots & & \cdots & 0 \end{bmatrix}_{30\times 30}$$

That is, $A_{\text{nonnormal}}$ is zero except for the entries 100 on the superdiagonal. It is strictly upper triangular, so its only eigenvalue is 0. In both cases $A \in \mathbb{C}^{m\times m}$ with $m = 30$, and $q_1 = b/\|b\|_2$. Let n be the number of Arnoldi steps.

	Normal case	Nonnormal case
Matrix dimension m	30	30
Arnoldi steps n	10	1
Starting vector b	random vector	$(1, 0.01, 0, \ldots, 0)^T$
Number of Ritz values	10	1

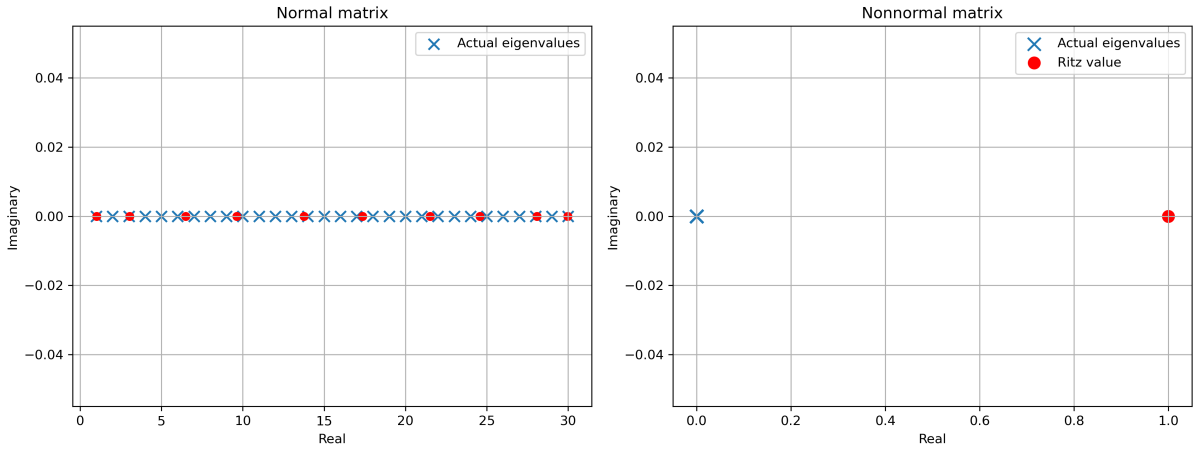


Figure 1: Actual eigenvalues and Ritz values for a normal matrix (left) and a nonnormal matrix (right).

Table 1: Normal matrix: Ritz values, residuals and distance to the spectrum.

Ritz value θ	Residual $\|Au - \theta u\|_2$	$d(\theta, \Lambda(A))$
1.006770	1.862×10^{-1}	6.770×10^{-3}
3.048372	8.135×10^{-1}	4.837×10^{-2}
6.485809	2.226×10^{0}	4.858×10^{-1}
9.626454	2.502×10^{0}	3.735×10^{-1}
29.977610	2.936×10^{-1}	2.239×10^{-2}
28.072269	1.062×10^{0}	7.227×10^{-2}
13.768214	3.405×10^{0}	2.318×10^{-1}
17.351938	3.714×10^{0}	3.519×10^{-1}
24.576341	1.994×10^{0}	4.237×10^{-1}
21.516959	2.973×10^{0}	4.830×10^{-1}

Table 2: Nonnormal matrix: Ritz value, residual and distance to the spectrum.

Ritz value θ	Residual $\|Au - \theta u\|_2$	$d(\theta, \Lambda(A))$
0.999900	9.999×10^{-3}	9.999×10^{-1}

Remark 4.1 (Breakdown). For this b we have $A_{\text{nonnormal}} b = e_1$ and $A^2_{\text{nonnormal}} b = 0$, so $K_2(A, b)$ is already A-invariant. Arnoldi breaks down at $n = 2$ and by Remark 4.1 the Ritz values are then exact eigenvalues, i.e. 0. So Table 2 is the case $n = 1$ only. It shows that a small residual does not imply that θ is close to the spectrum when A is nonnormal. It is a counterexample, not typical behaviour.

5 GMRES

We want to solve the linear system

$$Ax = b, \qquad A \in \mathbb{C}^{m \times m}.$$

For large A, computing $A^{-1}b$ directly is expensive in time and memory, and it does not use the sparsity of A. So instead we construct a sequence of approximations $x_0, x_1, x_2, \ldots$ such that $x_k \to x_*$, where $Ax_* = b$. GMRES (Generalized Minimal Residual method) is an iterative method of this kind.

Residual

For an approximation x_k the residual is

$$r_k = b - Ax_k.$$

Here $r_k = 0 \iff Ax_k = b$. So we choose the approximation which minimizes

$$\boxed{\|b - Ax_k\|_2}.$$

Krylov subspace

Start with an initial guess x_0 and let $r_0 = b - Ax_0$. We do not search over all of $\mathbb{C}^m$, only in

$$K_n(A, r_0) = \text{span}\{r_0, Ar_0, A^2 r_0, \ldots, A^{n-1} r_0\},$$

that is,

$$x_n = x_0 + z_n, \qquad z_n \in K_n(A, r_0).$$

Only products with A are needed. We never form powers of A or A^{-1}.

Arnoldi gives an orthonormal basis of this space. Take $q_1 = r_0/\|r_0\|_2$ and $Q_n = [q_1, \ldots, q_n]$. Every $z_n \in K_n(A, r_0)$ is $z_n = Q_n y$ for some $y \in \mathbb{C}^n$. So

$$\boxed{x_n = x_0 + Q_n y}.$$

Now we only need n coefficients instead of m, where $n \ll m$.

The least-squares problem

Proposition 5.1 (GMRES)**.** *Let $r_0 = b - Ax_0 \neq 0$, $\beta = \|r_0\|_2$, and suppose Arnoldi with $q_1 = r_0/\beta$ runs n steps without breakdown ($h_{j+1,j} \neq 0$ for $j = 1, \ldots, n$). Then $x_n = x_0 + Q_n y_n$ minimizes $\|b - Ax\|_2$ over $x \in x_0 + K_n(A, r_0)$ if and only if y_n solves*

$$\boxed{\min_{y \in \mathbb{C}^n} \|\beta e_1 - \tilde{H}_n y\|_2}. \tag{6}$$

Moreover $\|b - Ax_n\|_2 = \|\beta e_1 - \tilde{H}_n y_n\|_2$.

Proof. By Remark 2.1 the columns of Q_n form a basis of $K_n(A, r_0)$, so $y \mapsto x_0 + Q_n y$ is a bijection from $\mathbb{C}^n$ onto $x_0 + K_n(A, r_0)$. Substituting $x_n = x_0 + Q_n y$,

$$\|b - Ax_n\|_2 = \|b - A(x_0 + Q_n y)\|_2 = \|r_0 - AQ_n y\|_2.$$

By the Arnoldi relation (1), $AQ_n = Q_{n+1}\tilde{H}_n$, where $\tilde{H}_n \in \mathbb{C}^{(n+1)\times n}$. Also $r_0 = \beta q_1 = Q_{n+1}\beta e_1$, where $e_1 = (1, 0, \ldots, 0)^T \in \mathbb{C}^{n+1}$. Hence

$$\|r_0 - AQ_n y\|_2 = \|Q_{n+1}\beta e_1 - Q_{n+1}\tilde{H}_n y\|_2 = \|Q_{n+1}(\beta e_1 - \tilde{H}_n y)\|_2.$$

Since the columns of Q_{n+1} are orthonormal, $Q_{n+1}^* Q_{n+1} = I_{n+1}$, so $\|Q_{n+1}u\|_2 = \|u\|_2$. Thus

$$\|b - Ax_n\|_2 = \|\beta e_1 - \tilde{H}_n y\|_2 \qquad \text{for every } y \in \mathbb{C}^n,$$

and minimizing the left side over x_n is the same as minimizing the right side over y. □

So the m-dimensional problem reduces to a small least-squares problem with an $(n+1) \times n$ Hessenberg matrix. We solve (6) using QR factorization or Givens rotations to get y_n, and then

$$x_n = x_0 + Q_n y_n.$$

We check the residual norm $\|\beta e_1 - \tilde{H}_n y_n\|_2$. If it is not small enough we do more Arnoldi steps (or restart GMRES).

Remark 5.1 (Breakdown in GMRES). Suppose $h_{n+1,n} = 0$ at step n with $h_{j+1,j} \neq 0$ for $j < n$, and A is nonsingular. Then $AQ_n = Q_n H_n$, so for every y

$$r_0 - AQ_n y = Q_n(\beta e_1 - H_n y), \qquad e_1 \in \mathbb{C}^n.$$

H_n is nonsingular: if $H_n y = 0$ then $AQ_n y = 0$, so $Q_n y = 0$, so $y = 0$. Taking $y_n = \beta H_n^{-1} e_1$ gives zero residual, i.e. x_n is the exact solution of $Ax = b$.

The whole procedure is given in Algorithm 2.

GMRES as a polynomial approximation problem

We want to solve $Ax = b$. Suppose our initial guess is x_0 with residual $r_0 = b - Ax_0$. Then GMRES looks for the approximate solution of the form

$$x_n = x_0 + z_n,$$

where the correction is restricted to the Krylov subspace

$$K_n(A, r_0) = \langle r_0, Ar_0, A^2 r_0, \ldots, A^{n-1} r_0 \rangle.$$

Algorithm 2 GMRES

Require: $A \in \mathbb{C}^{m\times m}$, $b \in \mathbb{C}^m$, initial guess x_0, maximum number of steps $n_{\max}$, tolerance ε

Ensure: approximate solution x_n

$r_0 \leftarrow b - Ax_0, \quad \beta \leftarrow \|r_0\|_2, \quad q_1 \leftarrow r_0/\beta$
$n \leftarrow n_{\max}$
for $j = 1, 2, \ldots, n_{\max}$ **do**
    $v \leftarrow Aq_j$
    for $i = 1, 2, \ldots, j$ **do**
        $h_{ij} \leftarrow q_i^* v$
        $v \leftarrow v - h_{ij} q_i$
    end for
    $h_{j+1,j} \leftarrow \|v\|_2$
    $y_j \leftarrow \arg\min_{y\in\mathbb{C}^j} \|\beta e_1 - \tilde{H}_j y\|_2$
    if $\|\beta e_1 - \tilde{H}_j y_j\|_2 \le \varepsilon$ **or** $h_{j+1,j} = 0$ **then**
        $n \leftarrow j$
        break
    end if
    $q_{j+1} \leftarrow v/h_{j+1,j}$
end for
$x_n \leftarrow x_0 + Q_n y_n$

So

$$z_n = c_0 r_0 + c_1 A r_0 + \cdots + c_{n-1} A^{n-1} r_0.$$

We define a polynomial

$$s_{n-1}(t) = c_0 + c_1 t + \cdots + c_{n-1} t^{n-1}.$$

Then

$$\boxed{z_n = s_{n-1}(A) r_0}$$

Now $r_n = b - Ax_n$. We substitute $x_n = x_0 + z_n$:

$$\begin{aligned} r_n &= b - A(x_0 + z_n) \\ &= r_0 - Az_n \\ &= \big[I - A s_{n-1}(A)\big] r_0. \end{aligned}$$

We define

$$\boxed{p_n(t) = 1 - t\, s_{n-1}(t)}$$

Then

$$\boxed{r_n = p_n(A) r_0}$$

(i) p_n has degree at most n.

(ii) $p_n(0) = 1$.

Conversely, let $p(t) = 1 + a_1 t + \cdots + a_n t^n$ be any polynomial with $p(0) = 1$. Then $s_{n-1}(t) = \dfrac{1 - p(t)}{t}$ is a polynomial of degree at most $n-1$ and $p = 1 - t s_{n-1}$. So the

polynomials p_n above are exactly the polynomials of degree at most n with $p(0) = 1$. Therefore the GMRES problem can be written as

$$\boxed{\min_{\substack{\deg p \le n \\ p(0)=1}} \|p(A)r_0\|_2}$$

Basically we want a polynomial p which minimizes $\|p(A)r_0\|_2$, where r_0 is the first residual. For $x_0 = 0$ we have $r_0 = b$, which is the form used by Trefethen and Bau: $\min \|p(A)b\|_2$.

Proposition 5.2. *Let A be normal, $A = V\Lambda V^*$ with V unitary and $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_m)$, and let $c = V^* r_0$. Then for every polynomial p*

$$\|p(A)r_0\|_2^2 = \sum_{i=1}^{m} |p(\lambda_i)|^2 |c_i|^2.$$

In particular

$$\boxed{\frac{\|r_n\|_2}{\|r_0\|_2} \le \inf_{\substack{\deg p \le n \\ p(0)=1}} \max_{\lambda \in \Lambda(A)} |p(\lambda)|}$$

Proof. We have $p(A) = Vp(\Lambda)V^*$, where $p(\Lambda) = \operatorname{diag}(p(\lambda_1), \ldots, p(\lambda_m))$. Since $r_0 = Vc$,

$$p(A)r_0 = Vp(\Lambda)V^*Vc = Vp(\Lambda)c.$$

V is unitary, so $\|p(A)r_0\|_2 = \|p(\Lambda)c\|_2$, which gives the formula. Also $\|r_0\|_2 = \|c\|_2$, so

$$\|p(A)r_0\|_2 \le \max_i |p(\lambda_i)| \, \|r_0\|_2.$$

r_n minimizes $\|p(A)r_0\|_2$ over all p with $\deg p \le n$, $p(0) = 1$. So $\|r_n\|_2 \le \|p(A)r_0\|_2 \le \max_i |p(\lambda_i)| \, \|r_0\|_2$ for every such p, and taking the infimum over p gives the bound. □

Remark 5.2. GMRES minimizes $\|p(A)r_0\|_2$. It does not minimize $\max_{\lambda \in \Lambda(A)} |p(\lambda)|$. The minimax problem only gives an upper bound for the GMRES residual, and only when A is normal.

The formula in Proposition 5.2 says that the residual depends on the values of p_n at the eigenvalues and on the components c_i of r_0 along the eigenvectors. GMRES tries to find a polynomial which suppresses these components.

Example 5.1. *Let A be normal with $\Lambda(A) = \{1, 2, 3\}$ and*

$$p(t) = \Big(1 - \frac{t}{1}\Big)\Big(1 - \frac{t}{2}\Big)\Big(1 - \frac{t}{3}\Big).$$

Then $\deg p = 3$, $p(0) = 1$ and $p(1) = p(2) = p(3) = 0$, so

$$p(A) = Vp(\Lambda)V^* = 0.$$

At step n, GMRES minimizes $\|p(A)r_0\|_2$ over all polynomials with $\deg p \le n$ and $p(0) = 1$. So we get a new minimization at every step, and the minimizer p_n is different for different n. When n increases the set of allowed polynomials becomes bigger. The polynomial p above, is allowed for every $n \ge 3$. Hence

$$\|r_3\|_2 = \|p_3(A)r_0\|_2 \le \|p(A)r_0\|_2 = 0,$$

i.e. GMRES reaches the exact solution in at most three steps. GMRES minimizes over all of p. By Proposition 5.2

$$\|p(A)r_0\|_2^2 = \sum_{i=1}^{3} |p(\lambda_i)|^2 |c_i|^2.$$

If r_0 has no component along the eigenvector of $\lambda = 3$ (i.e. $c_3 = 0$), then $p(t) = (1-t)(1-\frac{t}{2})$ is enough and $r_2 = 0$.

Write $p(t) = 1 - t\,s_2(t)$ with $\deg s_2 \le 2$. Then $p(A) = 0$ gives $A\,s_2(A) = I$, so $A^{-1} = s_2(A)$ and

$$x_* - x_0 = A^{-1}r_0 = s_2(A)r_0 = c_0 r_0 + c_1 A r_0 + c_2 A^2 r_0 \in K_3(A, r_0).$$

So the exact correction already lies in the 3-dimensional Krylov space, however large m is. In general, if A has only k distinct eigenvalues $\lambda_1, \dots, \lambda_k$, then $p(t) = \prod_{i=1}^{k}(1 - t/\lambda_i)$ has degree k and GMRES converges in at most k steps. If the spectrum has only a few distinct eigenvalues, a low degree polynomial is enough.

Example 5.2. *Let A be normal with all eigenvalues in the disk $|\lambda - \gamma| \le \rho$, where $\rho < |\gamma|$ (so 0 is outside the disk). Take $p_n(z) = (1 - z/\gamma)^n$. Then $p_n(0) = 1$ and for every eigenvalue λ*

$$|p_n(\lambda)| = \left|\frac{\gamma - \lambda}{\gamma}\right|^n \le \left(\frac{\rho}{|\gamma|}\right)^n.$$

By Proposition 5.2

$$\boxed{\frac{\|r_n\|_2}{\|r_0\|_2} \le \left(\frac{\rho}{|\gamma|}\right)^n}$$

This p_n is only a candidate polynomial, not necessarily the minimizer, but GMRES does at least as well. So eigenvalues clustered away from zero give geometric convergence.

Example 5.3. *Suppose some eigenvalues $\lambda_1, \dots, \lambda_k$ are close to 0. If p_n does not vary rapidly near 0, then $p_n(\lambda_i) \approx p_n(0) = 1$, so the terms $|p_n(\lambda_i)|^2|c_i|^2 \approx |c_i|^2$ are not suppressed. This is only intuition for slow convergence. Eigenvalues near zero do not automatically imply slow convergence; the other eigenvalues, the degree of p_n and the c_i also matter.*

Now let A be nonnormal.

Proposition 5.3. *Let $A = X\Lambda X^{-1}$ be diagonalizable. Then*

$$\boxed{\|p(A)\|_2 \le \kappa_2(X) \max_{\lambda \in \Lambda(A)} |p(\lambda)|}, \qquad \kappa_2(X) = \|X\|_2 \|X^{-1}\|_2.$$

Hence

$$\frac{\|r_n\|_2}{\|r_0\|_2} \le \kappa_2(X) \inf_{\substack{\deg p \le n \\ p(0)=1}} \max_{\lambda \in \Lambda(A)} |p(\lambda)|.$$

Proof. $p(A) = Xp(\Lambda)X^{-1}$, so $\|p(A)\|_2 \le \|X\|_2\|X^{-1}\|_2\|p(\Lambda)\|_2$ and $\|p(\Lambda)\|_2 = \max_{\lambda\in\Lambda(A)} |p(\lambda)|$. The second inequality follows as in Proposition 5.2, using $\|p(A)r_0\|_2 \le \|p(A)\|_2\|r_0\|_2$. □

For normal A we can take X unitary, so $\kappa_2(X) = 1$. For highly nonnormal A the eigenvector matrix can be badly conditioned, $\kappa_2(X) \gg 1$, and then a polynomial which is small at every eigenvalue need not give a small $p(A)$. So for nonnormal matrices the eigenvalues alone can be misleading, and other tools (for example pseudospectra) are needed.

6 Numerical Experiment: GMRES Convergence

Disk versus rose spectrum

Let $m = 500$, let $G \in^{m\times m}$ have i.i.d. $N(0,1)$ entries and

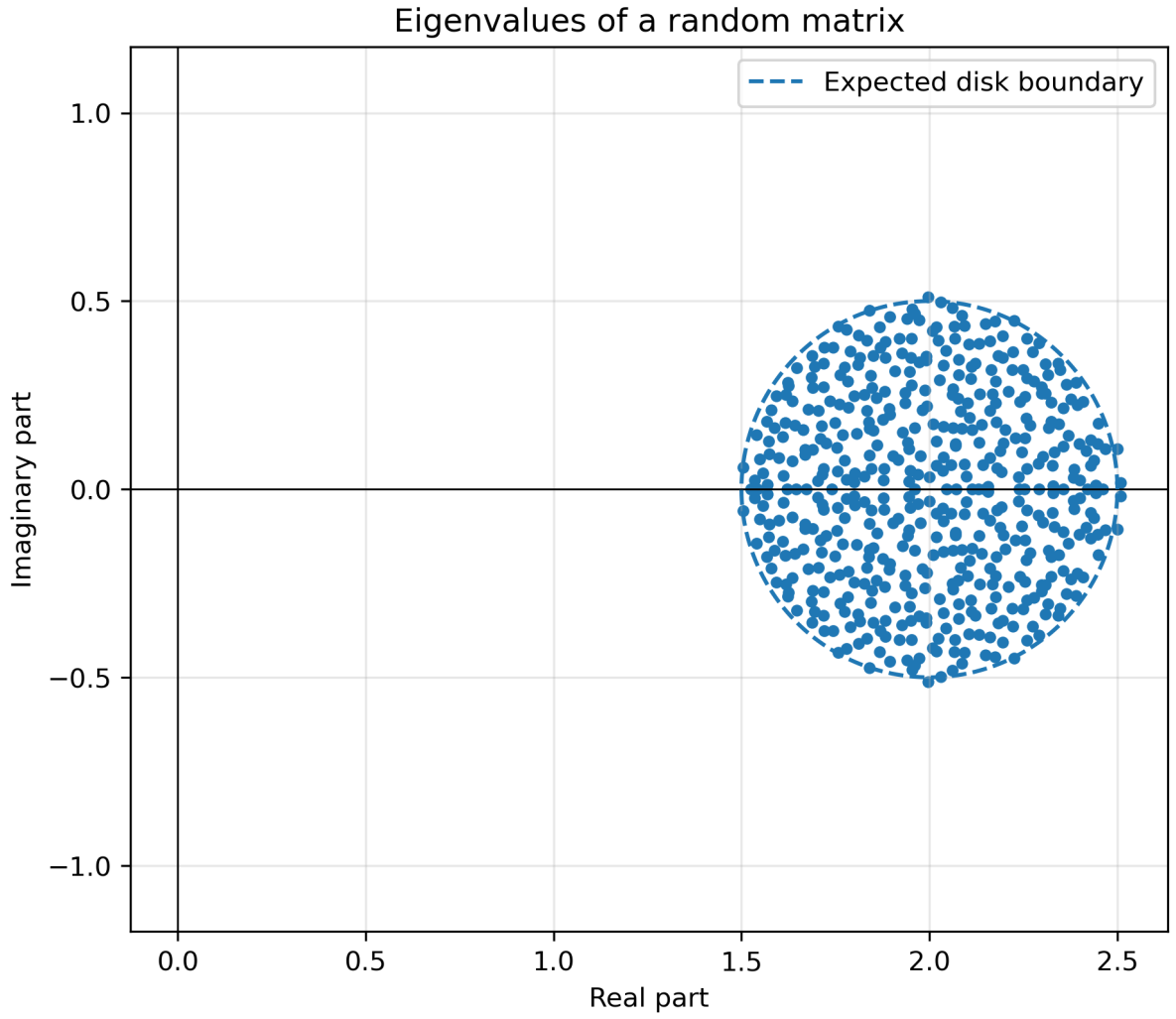
$$A = 2I + \frac{1}{2\sqrt{m}}\,G.$$

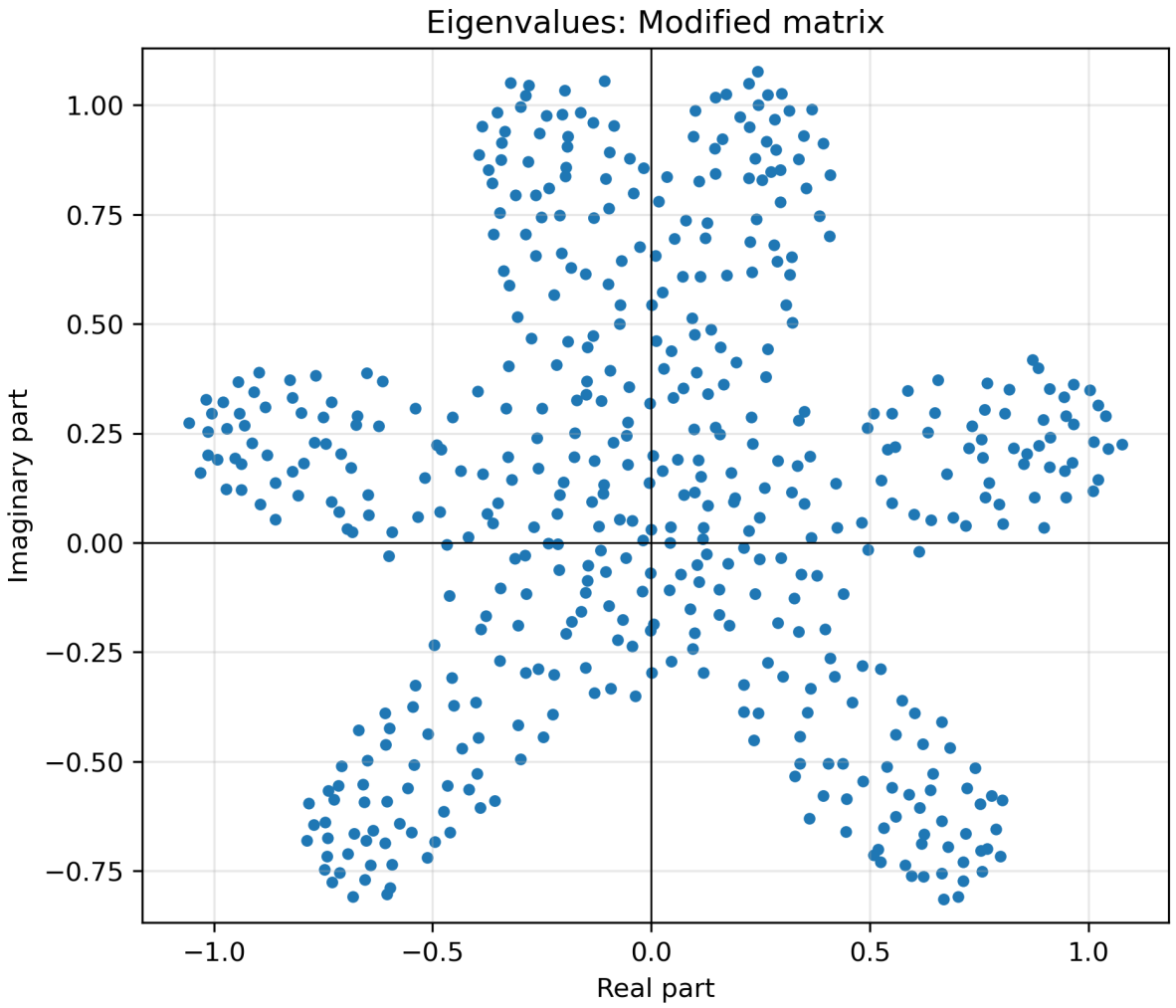
The eigenvalues of A fill approximately the disk $|\lambda - 2| \le \frac{1}{2}$, so $\gamma = 2$ and $\rho = \frac{1}{2}$ in Example 5.2. Let $k = 6$, $\theta_j = (j-1)\pi/m$ and

$$D = \operatorname{diag}(d_1, \dots, d_m), \qquad d_j = \big(-2 + \sin(k\theta_j)\sin\theta_j\big) + i\sin(k\theta_j)\cos\theta_j,$$

$$A_{\text{mod}} = A + D = \operatorname{diag}\big(i\sin(k\theta_j)e^{-i\theta_j}\big) + \frac{1}{2\sqrt{m}}\,G.$$

The diagonal part lies on a rose curve around the origin, so the eigenvalues of A_{mod} lie in a thickened rose that surrounds 0. We take $b = (1, \dots, 1)^T$, $x_0 = 0$ and run GMRES for $n = 1, \dots, m$.




Eigenvalues: Modified matrix
Imaginary part
Real part
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.0
−0.5
0.0
0.5
1.0

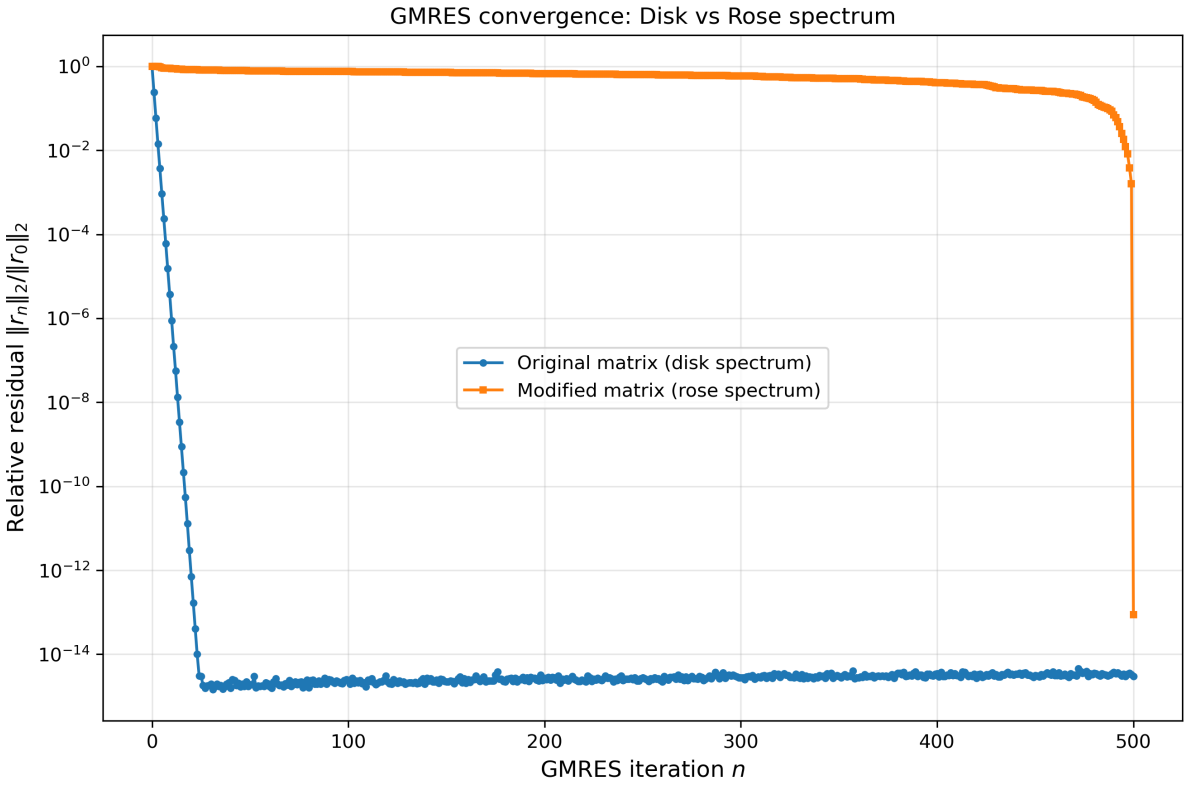
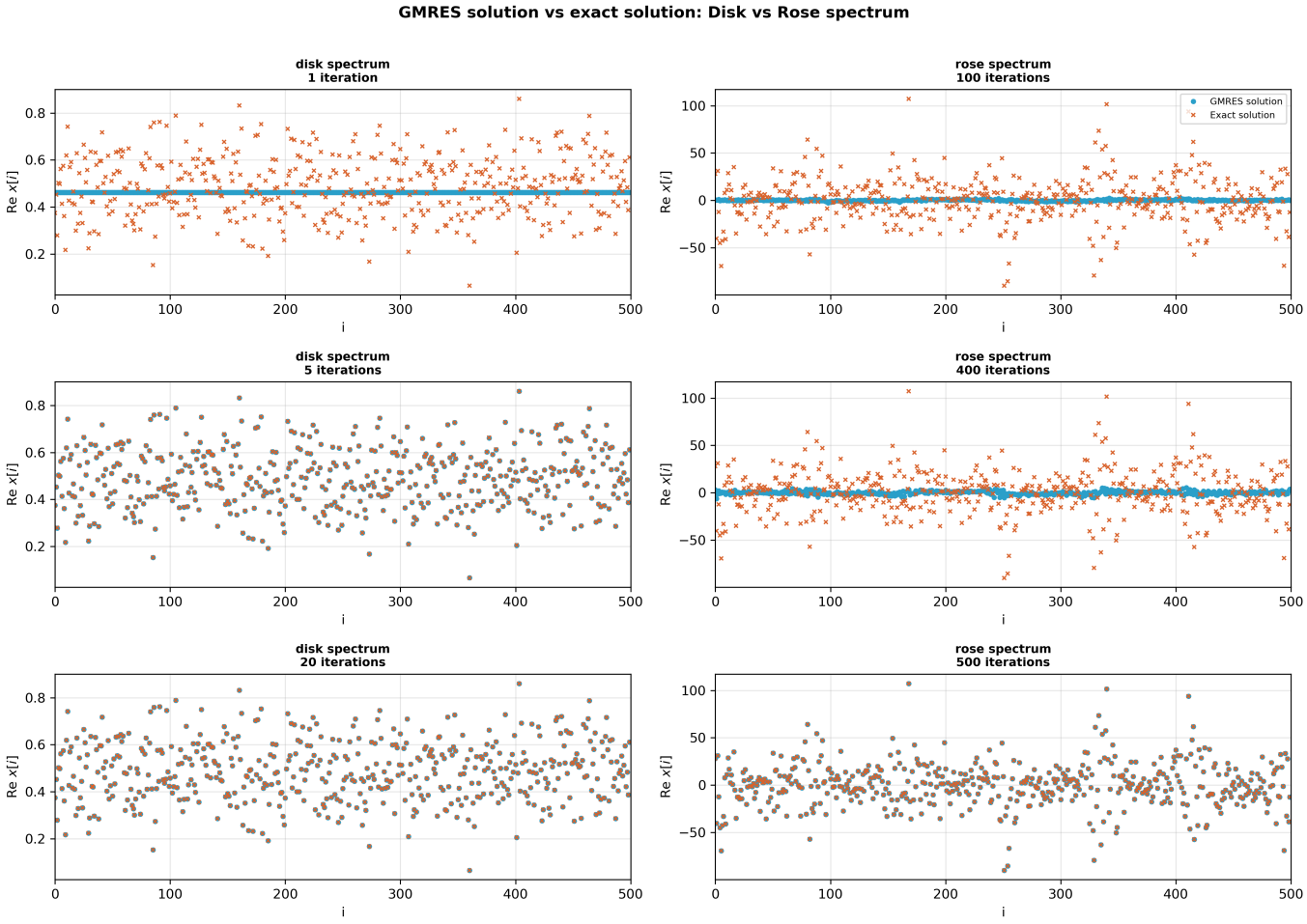


Figure 2: Top: eigenvalues of A (disk) and A_{mod} (rose). Bottom: relative GMRES residual $\|r_n\|_2/\|r_0\|_2$ for both matrices, with the reference 0.25^n.

Remark 6.1. A and A_{mod} are not normal, since they contain the random part G. For A, Proposition 5.3 with the candidate polynomial of Example 5.2 gives $\|r_n\|_2/\|r_0\|_2 \leq \kappa_2(X)\,(\rho/\gamma)^n$, with $\kappa_2(X) \approx 5 \times 10^2$ for unit-norm eigenvectors. For A_{mod} the eigenvalues surround 0, so no such bound is available.


GMRES solution vs exact solution: Disk vs Rose spectrum
disk spectrum
1 iteration
rose spectrum
100 iterations
GMRES solution
Exact solution
disk spectrum
5 iterations
rose spectrum
400 iterations
disk spectrum
20 iterations
rose spectrum
500 iterations
Re x[i]
i

7 Bibliography

References

[1] L. N. Trefethen and D. Bau III, *Numerical Linear Algebra*, SIAM, Philadelphia, 1997.

[2] D. S. Watkins, *Fundamentals of Matrix Computations*, 3rd ed., Wiley, 2010.

[3] G. H. Golub and C. F. Van Loan, *Matrix Computations*, 4th ed., Johns Hopkins University Press, Baltimore, 2013.

[4] A. Greenbaum, *Iterative Methods for Solving Linear Systems*, SIAM, Philadelphia, 1997.